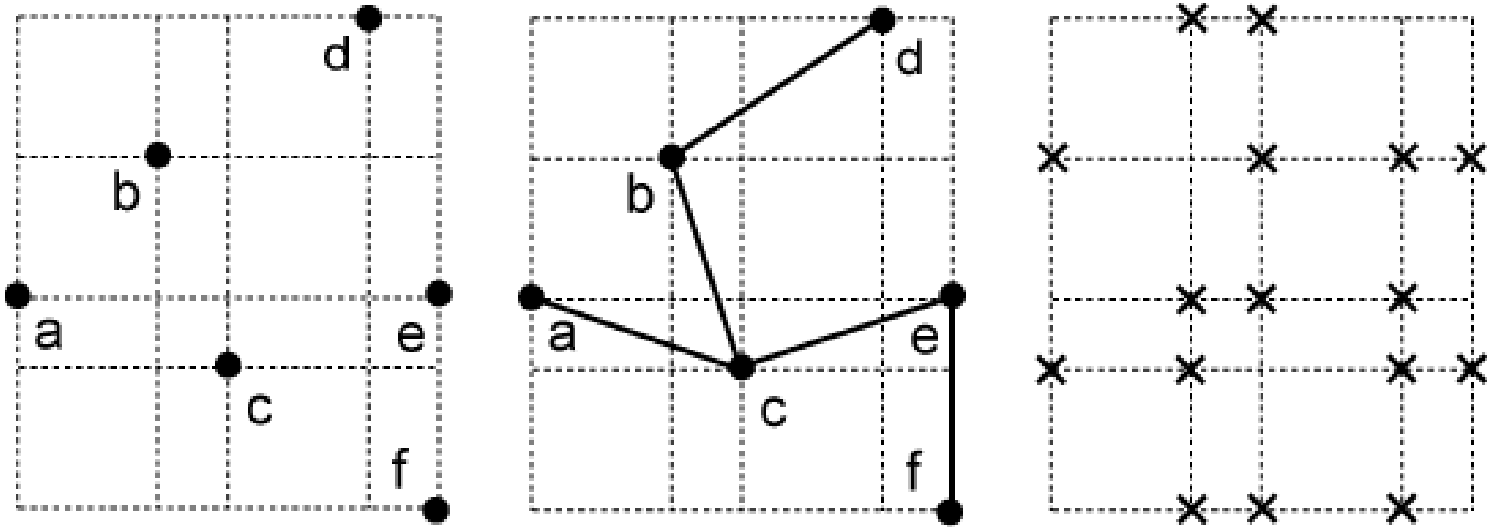
Kahng/Robins' Algorithm

Iterative 1-Steiner Insertion Algorithm

- Keep adding 1-Steiner point one-by-one until no more gain
- Find a 1-Steiner point by constructing a new MST on n+1 points for each element in the Stiner candidate set, and pick the candidate which results in the shortest MST
- Time complexity O(n3)

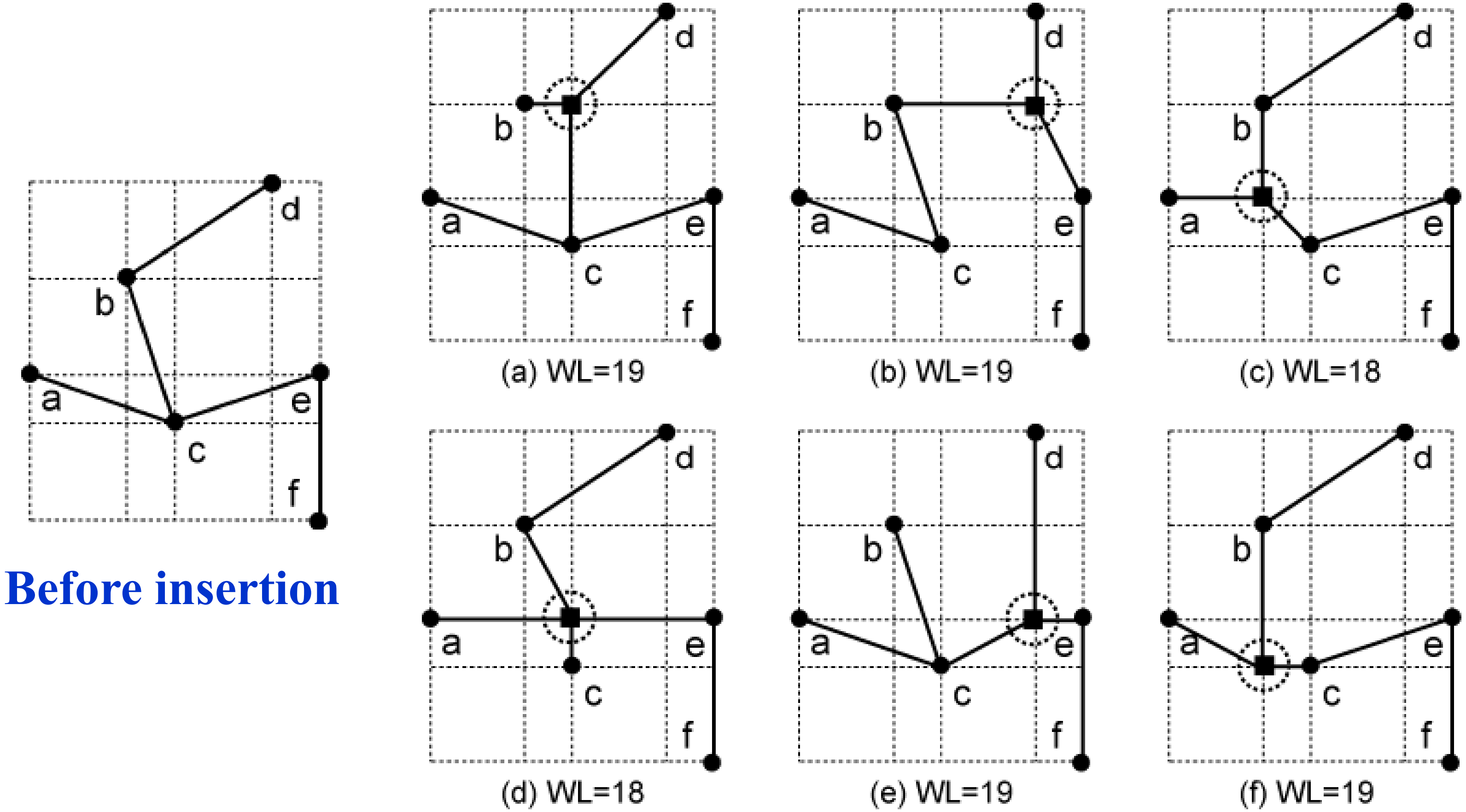
- Perform 1-Steiner Routing by Kahng/Robins
  - Need an initial MST: wirelength is 20
  - 16 locations for Steiner points



(a) (b) WL=20 (c)

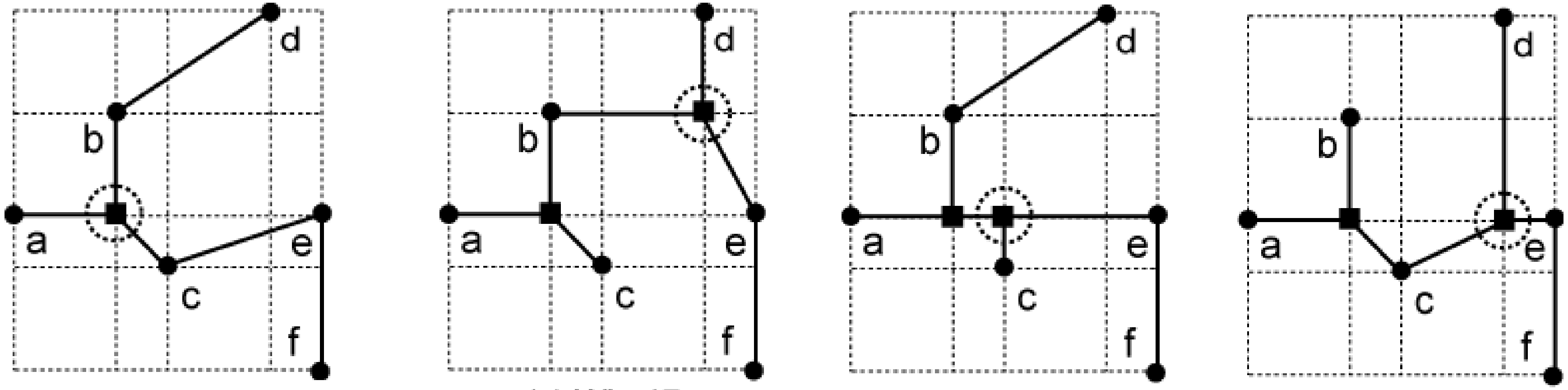
First 1-Steiner Point Insertion

- Total of six 1-steiner point candidates, pick (c) or (d)



Before insertion

- Second 1-Steiner Point Insertion
  - Total of three 1-steiner point candidates
  - Need to break tie again, note that (a) and (b) do not contain any more 1-Steiner point: so we choose (c)

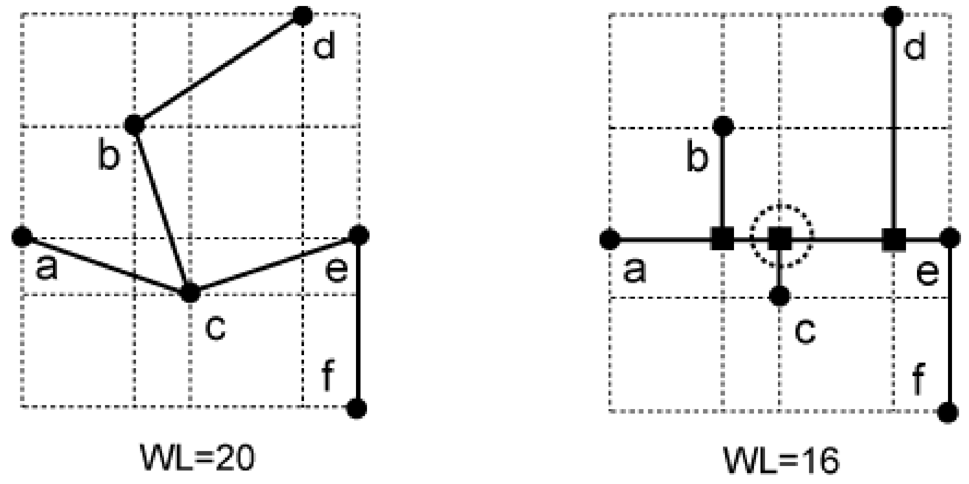


Before insertion (a) WL=17 (b) WL=17 (c) WL=17

Kahng/Robins' Algorithm Example

❑ Third 1-Steiner Point Insertion

- Tree completed: all edges are rectilinearized
- Overall wirelength reduction = 20 − 16 = 4



Before **After**